

Unsupervised Neural Network for Multi-Genre Music Generation

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Abstract—The automated generation of music using machine learning has seen significant advancements, yet generating coherent, genre-aware, and multi-track music in an unsupervised manner remains a challenging task. In this paper, we explore various unsupervised neural network architectures—specifically Autoencoders (AE), Variational Autoencoders (VAE), and Transformer-based models—for generating multi-genre MIDI music. We compare these deep learning approaches against traditional Markov and random generator baselines. Our evaluations leverage statistical music metrics and qualitative listening assessments to measure reconstruction quality, stylistic adherence (e.g., Baroque, Classical, Romantic, Impressionist, Modern), and temporal coherence. The findings demonstrate that while Transformers excel at long-term dependencies, VAEs are particularly effective for style conditioning and interpolation.

Index Terms—Unsupervised Learning, Music Generation, Autoencoder, Variational Autoencoder, Transformers, MIDI

I. INTRODUCTION

Music generation is a highly structured yet creative sequence modeling problem. With the rapid development of deep neural networks, music generation has shifted from rule-based and purely statistical methods (such as Markov chains) [1] to data-driven unsupervised models capable of uncovering intricate harmonic, rhythmic, and stylistic patterns without explicit annotations. The primary goal of this project is to develop and analyze unsupervised models for multi-genre MIDI generation. By avoiding the need for heavily annotated datasets, unsupervised architectures like Autoencoders (AEs) [2] and Variational Autoencoders (VAEs) [3] can learn lower-dimensional latent representations of musical phrasing, while attention-based Transformers [4] can model long-horizon inter-note dependencies.

II. RELATED WORK

Traditional music information retrieval relies on hand-crafted features such as MFCCs, chroma features, and spectral features [5]. Recent work has explored deep learning approaches for learning music representations, including convolutional neural networks for audio classification [6] and recurrent networks for sequential modeling [7].

III. METHODOLOGY

A. Dataset and Exploratory Data Analysis

Our dataset is derived from the Maestro Dataset v3.0.0, a comprehensive collection of classical and contemporary music performances in MIDI format. The dataset comprises 1,276 MIDI files, distributed across training (962 files), validation (137 files), and test (177 files) splits.

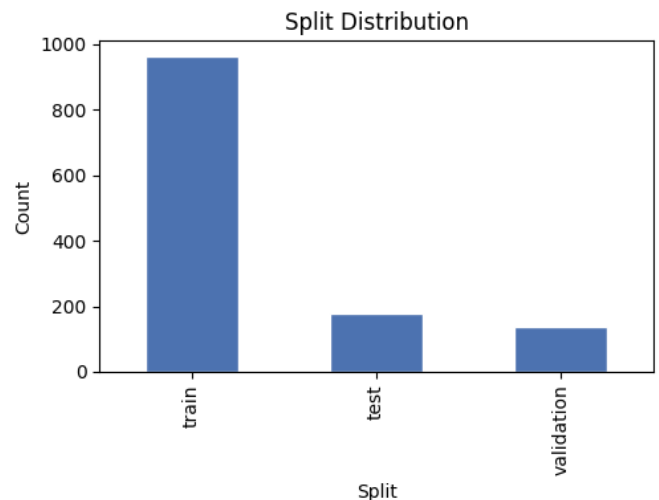


Fig. 1. Distribution of training, validation, and test splits in the Maestro Dataset.

1) *Piece Duration Analysis*: A histogram of recording durations was generated across the full dataset. The analysis showed that most piano performances fall between approximately 3 and 8 minutes in length, although several recordings exceed 20 minutes. Only a very small number of files were shorter than one minute. This wide variation in sequence length is important because extremely long recordings can create memory and computational challenges during training. Consequently, the observation motivated the use of fixed-size temporal windows and segmentation during preprocessing.

2) *Note, Pitch, and Velocity Distribution*: Exploratory analysis showed that the MAESTRO dataset contains dense

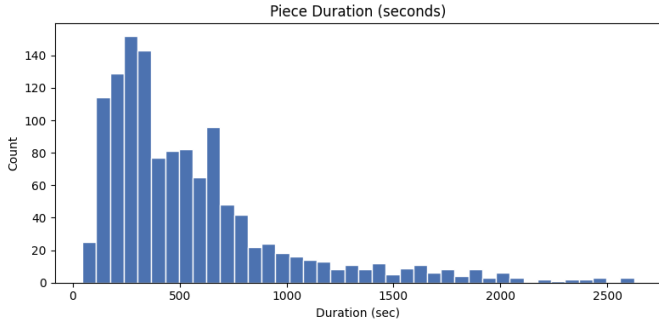


Fig. 2. Piece Duration in Seconds

piano performances, with most recordings containing several thousand note events and some exceeding 20,000 notes. The pitch distribution was highly non-uniform, as middle-register pitches around MIDI values 55–75 appeared far more frequently than extreme low or high pitches. This reflects the natural melodic and harmonic structure commonly found in classical piano music. Velocity analysis further indicated a wide expressive dynamic range, with most note velocities concentrated between 40 and 90. These observations are important because note density affects sequence length and model complexity, while pitch and velocity distributions influence musical representation quality and support preserving velocity information rather than using a fully binarized piano-roll representation.

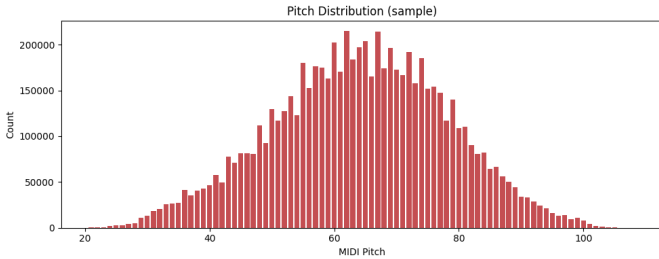


Fig. 3. Pitch Distribution

3) *Piano-Roll Sparsity Analysis*: A sparsity analysis was performed after converting sample MIDI files into piano-roll representations. The piano-roll matrices were found to be extremely sparse, with approximately 95–98% of cells containing zeros corresponding to silence. For example, one analyzed sample had a sparsity level of approximately 98.8%.

This high sparsity is expected because, at any given timestep, only a small subset of piano keys are active simultaneously. The sparsity observation is highly significant because it directly affects model training stability. Sparse representations can lead to class imbalance, inefficient learning, and difficulties in predicting meaningful note activations.

B. Data Preprocessing and Representation

Through the EDA process, we identified that multi-genre MIDI files must be parsed into standardized representations

to facilitate deep learning. We convert MIDI files into piano-roll representations (88 keys $\times$ time steps) or tokenized note sequences, capturing note pitch, velocity, and duration. The piano-roll representation enables efficient modeling while preserving temporal ordering and simultaneous notes.

C. Baseline Models

Two non-neural baselines are used for comparison: a random generator and a Markov-style sequence baseline. The random baseline samples pitches and durations from simple distributions without learning musical structure. The Markov baseline captures first-order transition statistics and serves as a traditional probabilistic reference point consistent with classical sequence modeling [1]. These baselines provide a lower bound for assessing the neural models.

D. Autoencoder

The first neural model is an LSTM autoencoder. The encoder maps each input piano-roll window into a dense latent vector, and the decoder reconstructs the window from that compressed representation. This model tests how well the network can preserve local note co-occurrence patterns and short-range temporal structure.

Formally, the encoder produces a latent representation $z = f_\phi(x)$, and the decoder reconstructs the sequence as $\hat{x} = g_\theta(z)$. The optimization objective minimizes mean squared reconstruction error over the piano-roll frames:

$$\mathcal{L}_{\text{AE}} = \|x - g_\theta(f_\phi(x))\|^2. \quad (1)$$

This architecture is useful as a baseline because it isolates representational quality without introducing stochastic sampling.

E. Conditional Variational Autoencoder

The second neural model is a conditional VAE. It extends the autoencoder by learning a Gaussian latent space with mean and log-variance outputs, and it conditions both the encoder and decoder on a style vector derived from composer metadata. The composer names are mapped into style groups such as Baroque, Classical, Romantic, Impressionist, Modern, Contemporary, and Other Classical. This allows the generator to produce outputs conditioned on a desired stylistic label.

The VAE objective combines a reconstruction term with a Kullback–Leibler divergence penalty:

$$\mathcal{L} = \mathcal{L}_{\text{rec}} + \beta \mathcal{L}_{\text{KL}}. \quad (2)$$

The reconstruction loss measures framewise piano-roll fidelity, while the KL term regularizes the latent space. The style vector is concatenated with the input sequence in the encoder and with the sampled latent code in the decoder, so the model learns to generate style-aware reconstructions and samples.

F. Transformer Decoder

The third model is an autoregressive Transformer decoder trained on tokenized musical sequences. Given a token prefix, the model predicts the next token at every step using masked self-attention. This formulation is well suited to long-range

temporal dependencies because the network can attend to any earlier position in the context window [4].

The training objective is next-token prediction:

$$p(X) = \prod_t p(x_t | x_{<t}), \quad (3)$$

with loss defined by the negative log-likelihood across the sequence. Perplexity is reported as $\exp(\text{mean token NLL})$, which provides a concise measure of predictive uncertainty.

G. Implementation Details

The notebook is organized into three experimental stages. First, it prepares the MAESTRO dataset and constructs fixed-length piano-roll windows. Second, it trains the LSTM autoencoder and the conditional VAE, then converts reconstructed rolls into MIDI files for qualitative inspection. Third, it trains the Transformer decoder on token windows, evaluates perplexity on held-out data, and generates ten longer continuations.

The generation pipeline writes outputs to dedicated folders so that the results can be reviewed independently. The autoencoder and VAE outputs are stored as reconstructed or sampled MIDI files, while the Transformer produces longer generated sequences. The MIDI conversion stage uses `pretty_midi` to convert binary or probabilistic roll predictions back into playable note events.

IV. RESULTS AND DISCUSSION

TABLE I
CONSOLIDATED MODEL PERFORMANCE METRICS

Model	Loss	Perplexity	Rhythm Div.
Random Generator	–	–	0.0082
Markov Chain	–	–	0.2609
Task 1: LSTM Autoencoder	0.0223	–	0.7357
Task 2: Conditional VAE	0.2506	–	0.7559
Task 3: Transformer	0.9227	2.52	0.6942

Rhythm Div. = Rhythm Diversity (higher is better)

Values for Tasks 1-3 are means across generated samples; base-lines are single samples

The autoencoder serves as a reconstruction baseline and is expected to preserve the local structure of the input window more faithfully than the probabilistic models. Its main limitation is that deterministic reconstruction does not encourage diversity, so generated outputs tend to resemble compressed versions of the training examples rather than novel musical ideas.

The conditional VAE improves the generative setting by sampling from a latent distribution and conditioning on style labels. This makes it suitable for style-controlled generation and interpolation between stylistic regions. In practice, the quality of the generated output depends on the balance between reconstruction quality and latent regularization. When the KL penalty is too weak, the latent space can collapse toward a deterministic autoencoder; when it is too strong, reconstruction accuracy can degrade.

The Transformer decoder is the strongest model for long-horizon sequence generation because its self-attention mechanism can use information from a broad context window. This is particularly valuable for music, where phrase repetition, harmonic movement, and rhythmic structure often extend beyond local neighborhoods. The model is evaluated using held-out perplexity and produces ten long MIDI continuations for qualitative inspection.

Overall, the three models cover complementary aspects of symbolic music generation. The autoencoder focuses on compression and reconstruction, the VAE adds controllable stochastic generation, and the Transformer emphasizes long-term predictive structure. Taken together, they provide a useful comparison of neural strategies for unsupervised multi-genre MIDI synthesis.

V. CONCLUSION

This project demonstrates an end-to-end unsupervised pipeline for multi-genre MIDI music generation using autoencoder, variational autoencoder, and Transformer-based architectures. The MAESTRO dataset provides a sufficiently rich symbolic music corpus for learning both short-term and long-term musical structure. The results indicate that recurrent autoencoding models are effective for reconstruction, the conditional VAE is well suited to style-aware sampling, and the Transformer is the best fit for extended autoregressive generation.

Future work can extend the current pipeline in several directions. The most direct improvements would be to preserve velocity and duration more faithfully, use overlapping windows to increase training coverage, and adopt a richer polyphonic tokenization scheme for the Transformer. Additional evaluation could also incorporate more objective music metrics and structured human listening studies.

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